



UNIVERSITY OF
PLYMOUTH

PUSL3190 Computing Project

Project Proposal

MidConnect - Digital Midwife Activity Tracking
System

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Chapter 01 – Problem Statement

The midwives have had a significant role in the Sri Lankan public health sector especially in the delivery of maternal and child health care. They are tasked with prenatal and postnatal clinic visits, child vaccination, home visits, parental counselling and community sensitization. However, documentation, and reporting procedures of such activities are still based on manual and paper-based approaches. Such conventionality has a number of inefficiencies and administrative problems.

Among the problems, one of them is that it is hard to track and monitor the daily tasks and performance of midwives. Records are managed manually, hence, retrieving, verifying or analysing them is labour intensive and prone to human error since records have been maintained in physical logbooks. Moreover, the paper-based systems are prone to losing, duplicating and degrading their records and hence, the quality of maternal and child health information is undermined. This also postpones reporting process to the higher administrative levels especially to the offices of Medical Officer of Health (MOH).

Furthermore, it is not easy to assess the productivity of midwives, determine areas of improvement, or produce timely reports without a centralized digital system that will help the MOH officials to achieve their goals. Such a deficiency at the digital level prevents proper decision-making, compromises accountability, and eventually seeks to influence the quality and efficiency of healthcare provision within communities. In a world where most of the administrative and clinical functions are increasingly going digital, there is a big gap in the midwifery sector, which is to be bridged by introduction of such systems.

The proposed project will overcome these issues by establishing a Digital Midwife Activity Tracking System (MidConnect) the web-based solution which will digitalize the daily activity management and reporting of the midwives, improve accuracy, accountability, and efficiency in the monitoring of the state of the population health.

Chapter 02- Project description.

Project Overview

The Digital Midwife Activity Tracking System (MidConnect) is a web-based application that aims at digitalizing and automating the process of activity tracking and reporting of the midwives who work within the context of the Sri Lankan public health framework. The system will enable the midwives to document their daily activities including home visits, vaccinations, clinic visits, and counselling sessions within an electronic setting. The data that is obtained will be stored safely in an online database that can be accessed and analysed by the respective MOH offices.

The design of the system will focus on user-friendliness, data integrity and availability. Midwives will report on a daily basis with the platform, and MOH administrators will be able to see summarized information, create analytical reports and track field performance. This system will also be very useful in improving the efficiency of the operations and the accuracy of the decision-making process since manual paperwork will be removed.

Key Functional Modules

1. User Authentication and Role Management- secure access to prevent unauthorized logins of midwives and MOH administrators with varied access permissions.
2. Daily Activity Logging- online forms to capture the daily activities like home visits, vaccination sessions and health education activities.
3. performance tracking dashboard - graphical data about the actions of every midwife, her attendance, and coverage of the services.
4. Automated Report Generation - summary and detailed reports would be automatically generated to be submitted weekly or monthly.
5. Data Analytics and Visualization- visual charts of the trends in maternal and child healthcare service delivery.
6. Program Scheduling and Reminders – calendar-based schedule tool of the events or health campaigns.

Project Objectives

- Objectives 1: To create and establish a digital platform that can be centralised to record the daily activities of midwives.
- Objectives 2: To enhance the accuracy and accessibility of data with the help of the online database management system.
- Objectives 3: To empower performance tracking and reporting in real-time by MOH officials.
- Objectives 4: To lessen the amount of administrative time and paper waste in the healthcare documentation procedures.
- Objectives 5: To improve delivery of maternal and child healthcare services in terms of monitoring and analysis.

Expected Outcomes

- A web-based application that is user friendly and replaces the currently used manual system.
- Better timeliness and accuracy of data regarding the activity of midwives.
- More effective visibility of MOH supervisors.
- More accountability and the use of data in the public health sector.

Project Keywords

- Digital Health System
- Activity Tracking
- Maternal and Child Health
- Public Health Monitoring

Chapter 03 – Research Gap

The digital transformation of healthcare systems has been a major trend across the world, and there is also a clear gap in the uptake of the said solutions into the primary healthcare setup in Sri Lanka. The vast majority of digital health systems installed in the country, including the Health Management Information System (HMIS) and the District Health Information Software 2 (DHIS2), are not designed to support the community-level service delivery but focus on managing hospital and clinical data (Manoj, 2012; Peiris et al., 2023). Despite these efforts enhancing the flow of institutional data, they do not sufficiently represent the operations and workload of field-level health workers including public health midwives (PHMs) (De Silva et al., 2020; Rodrigo et al., 2013).

The current mobile-based health applications (mHealth) like e-Praja Suwa Arunalu have proposed novel models of digital record-keeping and sharing of information. Nevertheless, they continue to rely on smaller volumes of manual data entry and are seldom connected to central analytics applications where real-time tracking and performance assessment could be conducted (Rodrigo et al., 2013; Lwin et al., 2016). Moreover, home visits, immunization tracking, maternal counselling, and clinic attendance, which are the daily tasks performed by midwives, are still documented with the help of traditional logbooks and then manually transferred to the offices of MOH, making the process slow and leading to discrepancies in data (Vithana et al., 2022; Senanayake et al., 2023).

Meanwhile, the digital health infrastructure in Sri Lanka has been focused on information exchange in the curative sector but not on preventive and community-based service provision. Electronic hospital systems, e.g., HHIMS, and the Master Patient Index (MPI) are the primary focus of patient-centric interoperability and clinical record integration (Jayathissa, Dissanayake and Hewapathirana, 2018; Jayathissa and Hewapathirana, 2023). Even though such systems are technologically viable, they lack customized providers of performance assessment, accountability, and planning of the activities of field health workers. The current gap, therefore, lies in bridging the gap between the grassroots and the sector of healthcare workers and the ecosystem of national health information (Wijesinghe et al., 2025).

It is against this backdrop that the proposed MidConnect -Digital Midwife Activity Tracking System is proposed to fill this gap by developing a single, intelligent, and community-oriented digital platform on which midwife activities in Sri Lanka can be reported, analyzed, and tracked in real-time. The next subsections indicate the main gaps in research and application that were detected in this regard.

3.1 Technology Gap

Even though Sri Lanka has implemented a number of e-health platforms, the majority of them are hospital-specific systems with a focus on patient data and reporting of clinical data. Current technology fails to provide a dynamic data visualization and geo-map, or smart analytics to monitor maternal and child health on a community-level (Rodrigo et al., 2013; Peiris et al., 2023). Also, there is still a lack of interoperability between current systems, including DHIS2, HHIMS, and MPI, which leads to data silos (Jayathissa and Hewapathirana, 2023).

MidConnect closes this technological gap due to the usage of an integrated database model and automated analytics modules targeting specifically the midwife workflow. It presents the options of

real-time dashboards, automated reports generation, and cloud-based data synchronization-technologies not introduced yet in the present public health information architecture of Sri Lanka.

3.2 Methodology Gap

Most of the past studies and systems have utilized quantitative data-entry methods without implementing any intelligent decision-support or feedback solutions to the end consumers (Manoj, 2012; Lwin et al., 2016). These systems are usually more concerned with data gathering and not data interpretation, hence they leave the administrators with raw data which do not provide actionable information. Besides, there are not many projects that have used participatory design approaches that involve the participation of real midwives in the development and testing (De Silva et al., 2020).

MidConnect suggests a hybrid approach to methodology, which is an integrated user-centred design, iterative-prototyping, and agile development practice. This guarantees the feedback provided by midwives, MOH officers and supervisors at all times hence aligning the technological solution with the actual operational requirements. This kind of approach facilitates usability and sustainability to resolve a long-standing methodological gap in previous health-information-system projects.

3.3 Contextual Gap

The contextual constraints of the current digital-health systems are mainly connected with their unwillingness to be localized and adjusted to the specifics of healthcare hierarchy in Sri Lanka and cultural background. Alternatively, systems made in foreign nations are usually imported with little adaptation to local organizational systems, linguistic inclinations, and report styles (Senanayake et al., 2023; Vithana et al., 2022). Moreover, unlike the majority of the digital initiatives targeting tertiary hospitals, little is thought about the rural and semi-urban operational setting of midwives (De Silva et al., 2020).

MidConnect tries to address this contextual void by offering a system tailored to the public-health midwifery network in Sri Lanka. It is consistent with the reporting standard of MOH offices, bilingual interface, and remote offline data entry can be adopted. Such a localized strategy is essential to be inclusive and practical so that digital health transformation can be spread beyond the hospital to the community level where preventive care is provided.

Chapter 04 – Requirements Analysis.

Functional Requirements

1. Authentication and Access Control User authentication and role-based permissions Each user (midwife, supervisor, MOH officer) should be authenticated with secure credentials.
2. Activity Recording Module: Midwives should have the ability to record the information about home visits, clinic visits and immunization programs.
3. Dashboard and Analytics: The system will be required to provide real-time dashboards showing coverage statistics and performance indicators.
4. Automated Report Generation: Report Periodic reports on the activity should be automated and available to export in different forms.
5. Notification System: Notifications and warnings about the next visit, time limits, or data mismatch.
6. Search and Filter Feature: The possibility to search data by region, date, midwife ID, or type of service.
7. Audit Trail: Document user activities to provide data integrity and auditability.
8. Backup and Recovery: Automatic backup of data to avoid the loss of information in case of failure of the system.

Non-Functional Requirements

1. Usability: The interface should also be user-friendly, easy to use and understand and accessible to users who are not very digital natives.
2. Performance: The system is to be efficient in the number of users who can use it simultaneously with the response time being minimal.
3. Security: Employ encryption and safe authentication software to conserve confidential health information.
4. Scalability: The system should be able to accommodate other features and more users.
5. Maintainability: The code should be well documented and modular towards future updates.
6. Availability: It will have 99% availability so that it is available at all times.
7. Portability: The system must work across numerous devices such as tablet, smart phones.

User Requirements

1. Midwives need a user-friendly system that will help them to ease their reporting load.
2. MOH administrators should have real-time access to midwife activities and coverage statistics.
3. The supervisors need access to performance dashboards and alerts to low-activity areas.
4. User management tools, data backup tools and troubleshooting tools are required by system administrators.

Technical Requirements

- Front-end: HTML, CSS, JavaScript.
- Back-end: PHP
- Database: MySQL
- Hosting Platform: XAMPP (local development), which can be deployed to a cloud server.
- Operating System: Windows.
- Hardware: Laptop or personal computer with a stable internet connection.

Knowledge and Skills Necessary

- PHP and MySQL Web development.
- Database specifications and normalization.
- Usability testing and front-end interface design.
- Creation of reports and visualization of data in the form of graphs and charts.
- General understanding of healthcare privacy and security standards.

Special Tools or Devices

It does not require any special hardware equipment besides a typical computer. However, the mobile compatibility will be taken into account to enable midwives to access the system through smartphones or tablets.

Chapter 05 – Finance (Budget Plan)

Item	Description	Estimated Cost (LKR)
Domain and Hosting (1 year)	Online deployment	8,000
Development Tools / IDE	Visual Studio Code, XAMPP (Free)	0
Internet Connections	Development and Testing	3000
Operational	Printing, documentation, testing	4000

Total Estimated Cost $\approx$ 15,000 LKR

Tangible Benefits

1. Less Work Particular to the Administration: Removes the use of paper-based records, which saves time and energy among midwives.
2. Enhanced Data Accuracy: The human element of input validation is reduced through automated input.
3. Better Reporting Performance: Instantaneous reports as opposed to manual reports.
4. Increased Resource Usage: Allows the MOH offices to distribute resources and personnel.
5. Transparency in Operations: All performance tracking guarantees accountability on all service levels.

Intangible Benefits

1. Better Decision-Making: Real-time analytics boost strategic planning and tracking.
2. Employee Satisfaction: Midwives get a lower work load and better job satisfaction.
3. Public Trust: Digital trust enhances the confidence of the community on the systems of health.
4. Innovation Culture: Promotes the use of technology in government hospitals.
5. Sustainability: Lasting digitalization promotes environmentally-friendly, paperless processes.

Chapter 06 - External Organizations / Staff.

- External Party: Local Medical Officer of Health (MOH) office- to provide advice and knowledge about the real-life midwife practice.
- Supervisor: University academic supervisor (Mrs. Chathurma Wijesinghe) assigned - to offer technical and academic supervision.
- Supporting Staff: The chosen midwives or field officers of the public health that will help in the gathering of requirements and user-feedback throughout the testing.

Chapter 07 – Time Frame / Timeline

Activity Breakdown Table

Task	Predecessor	Duration (Weeks)	Colour in Gantt Chart
A - Project Initiation	-	2 weeks	Blue
B - Research & Requirement Gathering	A	3 weeks	Red
C - System Design	B	3 weeks	Green
D - Phase 1 Development (Frontend)	C	4 weeks	Yellow
E - Phase 2 Development (Backend)	D	4 weeks	Orange
F - Testing & Evaluation	E	3 weeks	Purple
G - Preparation of final report	F	2 weeks	Pink

Totally estimated time: around 21 weeks (5 months).

Click here to see the gantt chart -

<https://1drv.ms/x/c/036ab40ffd411d2a/EdhLudDhFqlFhw9z8D1wO28BcRuz6OIr0CGfHdpIsza3TQ?e=WfGXXd>

Conclusion

The MidConnect - Digital Midwife Activity Tracking System suggests one new method of enhancing the services provided by the Sri Lankan maternal and child health sector by digitizing its services. The project will improve the efficiency, accuracy, and accountability of midwife operations by substituting manual reporting with paper-based reporting with a centralized digital system. Providing real-time monitoring and decision-making, the data-driven design of the system assists the Medical Officer of Health (MOH) offices in their work with the field and performance assessment. Finally, the project will help the country to

achieve the larger e-health objectives in the sense of ensuring sustainable, transparent, and smart healthcare management that will benefit the health professionals and their communities.

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